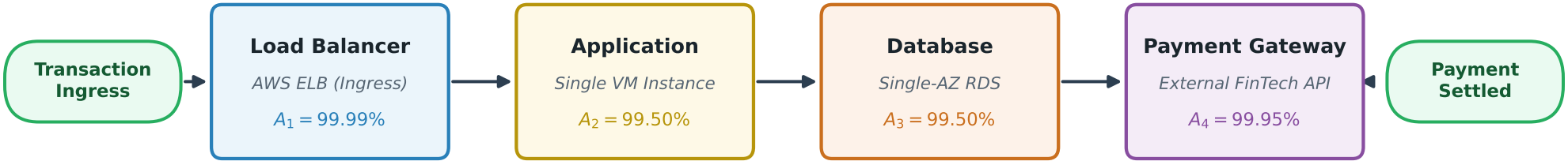
Scenario 1: Baseline Architecture (Pure Series RBD)

$$A_{\text{base}} = \prod_{i=1}^{4} A_i = A_{\text{LB}} \cdot A_{\text{App}} \cdot A_{\text{DB}} \cdot A_{\text{PGW}} = 98.943\% \implies 92.58 \text{ h/year downtime}$$



Critical Flaw: Every component is a Single Point of Failure (SPOF). App and DB alone account for >94% of total downtime.